SHADOW NETWORK: MOBILE APP-BASED PEER-TO-PEER DISASTER MESSAGING SYSTEM



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CHAPTER I

INTRODUCTION

Background of Study

The present situation with global disaster management demonstrates a fundamental lack of fail-safe mechanisms due to single point of failure which is reliance upon cell towers in both land-based and mobile technologies which becomes apparent when disaster strikes [1]. The implementation of improved technology in disaster response is an important step towards preventing devastation by disasters occurring in the first place [2]. The problem is very prevalent in the Philippines, particularly, where disasters such as typhoons and floods usually break down technologies before they reach full capacity [3]. The situation in Tagum City, where high-risk barangays are cut off from connection during calamities as the Libuganon and Hijo Rivers completely isolate them both physically and digitally [4].

Compliance with international frameworks drives this study's alignment with UN Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities (Target 11.5), which focuses on reducing disaster impacts on vulnerable populations [5]. This alignment is justified by the system's ability to preserve critical communications when cellular infrastructure fails during severe natural hazards [6]. Furthermore, integrating off-grid peer-to-peer (P2P) networking directly operationalizes local and national research priorities in Information and Communications Technology (ICT) and Disaster Risk Reduction and Management (DRRM) [7]. By enabling structured emergency payloads to bypass grid collapse, this study provides a scalable, decentralized mechanism to protect digitally isolated communities in the flood-prone Davao Region [8].

Barangays adjacent to the river banks in Tagum City are representative of a specific geography where technical issues pertaining to infrastructure dependency are solved using decentralized methodologies [9]. Data volatility is

still an important issue because conventional messaging lacks durability or resilience with respect to device reboots or battery issues [10]. Additionally, rescue coordination is, in many cases, limited due to the lack of structured geographical information from standard text messages needed by Tagum CDRRMO to prioritize victims [11]. Experimental estimation of the performance of message propagation under this river-locked environment corresponds to the last research gap in determining the level of effectiveness of communication technology in isolated last-mile areas [12].

The Shadow Network presents a technical solution for turning Android smartphones into continuous communication relays for isolated victims. Using dual protocol methods for neighbor identification, the system ensures the possibility of discovering neighboring devices with maximum energy efficiency in mobile environments [13]. However, the effectiveness of this multi-hop communication system relies solely on the willingness of users to engage with the process and profile of available hardware. Specifically, for a communication hop to take place, intermediate users have to turn on their Wi-Fi/Bluetooth radios and install the application in their devices.

The current research endeavor is in concrete synchrony with the research frameworks employed at the institutional, sectoral, and national levels with the objective of improving community disaster resilience. With respect to the institutional fiat, it supports the RDE Agenda (2026-2030) of the Davao del Norte State College (DNSC) under the umbrella of Community Resilience to the extent of implementing localized and ICT-supported responses to emergencies. On the sectoral front, the research also implements the CHED A.C.H.I.E.V.E Agenda (2025-2030) in a way that it fulfills the harmonized SDG-based R&D and Innovation agenda through the establishment of a strong communication system in times of disasters. The national level is addressed through the implementation of the DOST Harmonized National Research and Development Agenda (HNRDA 2022-2028) concerning the infrastructure, energy, and emerging technologies sector and focusing on the ICT Innovations in the field of disaster risk reduction and climate change adaptation.

Objectives of the Study

The primary goal of this study is to develop Shadow Network, a mobile peer-to-peer (P2P) messaging system that provides reliable, offline communication for selected communities in Tagum City that suffers total network collapses and during natural phenomena like floods. Specially, the project will:

1. Develop a shared or a community base P2P framework using Wi-Fi Direct and BLE to enable Android-to-Android SOS broadcasting without cell towers or internet.

2. Implement a Store-Carry-Forward (SCF) routing system to make sure that emergency records survive unsteady connectivity and spread reliably across physical network gaps.

3. 3Construct a GPS-driven geospatial tracking component using an organized location coordinates and accuracy data, allowing rescuers to visually track and pin a victim's exact physical location on an offline triage canvas.

4. Validate how the message travels efficiently across the last-mile gap by measuring delivery success rates and node-to-node wait time in selected communities.

Significance of Study

The findings of this study will show us on how to improve disaster response and emergency communication, specifically for regions prone to network system failure. The following sectors will benefit from this research:

- *Barangay Residents of Tagum City.* This study provides a phone application that helps android users in flood prone communities to call for help during a flood, even without cellular service. It shares the users exact location and needs like food or medicine when networks are down.

- *City Disaster Risk and Management Office (CDRMO).* The system helps rescuers by putting organized distress messages directly onto a map in an admin website for the rescuers and this removes messy messages confusion, so teams can rank and plan rescue mission further.

- *Local Government Units (LGUs).* This research offers a cheap software tool for disasters that works with an already existing technology. Because it doesn't need costly upgrades, local communities can easily keep using it.

- *The Field of Information Technology.* This study adds to what we know about offline mesh networks and how the Store-Carry-Forward (SCF) logic and P2P Mesh Networking works. It shows how local phone databases like SQLite, can save and protect information when there is no internet or electricity.

- *Future Researchers.* This project paves a way for future research on offline networks. Specifically, it helps find better ways to save phone battery and pass messages across multiple unrooted Android phones.

Scope and Limitation

Shadow Network is designed as a community shared, Android-based communication system to eliminate single points of failure in Tagum City's flood prone barangays using a resilient P2P mesh. Using Wi-Fi Direct and Bluetooth, it passes messages from phone to phone entirely offline. This offline capability is validated through simulation testing using android smartphones set only to Airplane mode during testing, through this the study can prove that phones can find and connect to each other automatically without cellular service. To keep data from getting lost, the system saves messages locally on the phone until the system can find a device to be passed along with the data.

This software only solution prevents network traffic congestion by using a data-filtering l ayer utilizing Jaccard Similarity and Levenshtein Distance algorithms automatically identifying and removing duplicate, near-duplicate, and routing identified spam messages into a designated spam category. While designed to

connect communities with little to no signal areas to responders, the system does not identify false human reports.

This system only works on Android. It does not support iPhones because Apple restricts offline mesh networks. The app runs on a closed local network, meaning it cannot access the internet, send standard texts, or open social media. To save battery and network space, the app blocks large files like photos and videos. Finally, users must preinstall the app and keep both Wi-Fi and Bluetooth turned on for the system to pass messages along.

Review of Related Literature and Works

When a disaster knocks out communication, emergency and rescue teams may find it hard to conduct help and rescue. This study shows how switching to local phone-to-phone networks can save lives. It focuses on keeping people connected during the critical first hours of an emergency, especially in high-risk areas in Tagum City where traditional phone networks may completely fail in the time of need.

Related Literatures

Vulnerabilities of Centralized Communication in Disasters

Previous studies show that dependence on centralized systems is one of the greatest risks in disaster response since even a single damaged cell tower can disable rescue communications for a whole area [14]. Previously, even though there have been advances in technology and telephones, they still remain primarily reliant on cell towers. When these towers fail in an emergency, victims cannot contact anyone for assistance [15]. This issue exists all over the world because emergency services are often overly reliant on weakened power lines which are usually the first to fall during any serious disaster [16].

Local Storage Persistence and Store-Carry-Forward (SCF) Architectures

In order to resolve central network breakdowns, scientists use mobile phones to ensure that data is secured, which leads to smartphones becoming persistent data storages instead of temporary transmitters [17]. The tested ideas in this field are in accordance with the Store-Carry-Forward (SCF) concept, which is a delay tolerant networking method in which information is kept at the user's physical location until some other connection is created [18]. These primary ideas are essential to ensure that disconnected communities are safe. A series of real-life tests show that the data is supposed to be sent at a local level and function under full network failures [19]. As a result, the utilization of local routing methods makes it possible even during offline mode for messages and emergency alerts to be transmitted via the same network.

Decentralized Mobile Ad-Hoc Mesh Networking

The results of the study bring forth two key lessons: firstly, networks can operate without towers and secondly, phones must perform local data storage. Simulations have shown that Wi-Fi and Bluetooth technology enables devices to connect with each other without routers, thus providing a possibility to set up emergency networks that operate off the grid [21]. Recent research has demonstrated that using mobile devices for carrying and transmitting information is especially effective in remote communities near rivers where the signal is obstructed by water and trees [22]. This trend has become necessary in the light of the fact that traditional social media and texting applications fail completely in case when cell towers are offline [23].

BLE Protocol Efficiency and Local Geographic Constraints

Tests prove that low-power Bluetooth is the best tool for disasters. It helps rescuers track victims safely without consuming phone batteries faster[24]. Local

studies prove that emergency communication technology only works if it is designed around the local landscape[25].These ideas also show that newer mesh systems work. They turn normal smartphones into an importance of a larger network that automatically fixes its own broken links [26]. Studies also show that for these networks to work, the most critical factors are how fast and how reliably phones can detect nearby devices[27].

Communication Gaps and Mobile Volatility

There is a gap in research on the fast-changing data of mobile mesh nodes in tropical river basins[28]. Recent studies show that hills and trees block wireless signals in remote countryside areas[29].Tests prove that saving data directly to an Android phone keeps it safe during long disasters. [30].

Related Works

Existing systems similar to the Shadow Network range from generalized mesh chat applications to specialized emergency frameworks. Foundational projects like the Serval Mesh have practically demonstrated the power of resilient, infrastructure-less platforms for crisis scenarios where traditional telecommunications are paralyzed [31]. Building on these principles, modern applications like IAWHINA have focused on the user-end experience, specifically designing offline mobile interfaces that function without any dependency on a centralized server or global internet connection [32]. Locally, the MO-DI-FI system has demonstrated the practical application of Wi-Fi Direct technology for search and rescue operations in the Philippines, providing real-time voice and location sharing for responders in areas without network connectivity [33].

To complement these frameworks, several international decentralized platforms serve as active empirical benchmarks. The commercial application Bridgefy establishes the real-world viability of utilizing Bluetooth Low Energy (BLE) protocols within multi-hop topologies to propagate data packages across dense

civilian populations during localized blackouts [34]. To address the data-integrity challenges of transport grid paralysis, the open-source architecture Briar provides a validated precedent for decentralized local database state synchronization, relying on secure Delay-Tolerant Network (DTN) routing protocols rather than centralized cloud architectures [35]. Furthermore, open-source community ecosystems like Meshtastic demonstrate the practical execution of routing structured off-grid spatial and telemetry data across ad-hoc nodes using managed flooding algorithms [36].

Despite these advancements, a critical gap remains in localized systems that combine these robust routing protocols with the specific operational needs of the Davao Region. Existing literature often overlooks the necessity of local data persistence on individual nodes to prevent SOS loss during power failure. This study addresses these specific technical voids by integrating SQLite-based storage with structured SOS messaging, providing a tailored dashboard for the Tagum CDRRMO that existing global and local frameworks currently lack..

Table 1. Comparison of Related Works

Features	**Shadow Network**	**Meshtastic**	**Briar**	**Bridgefy**	**Serval Mesh**	**IAWHINA**	**MO-DI-FI**
Offline P2P Messaging	✓	✓	✓	✓	✓	✓	✓
Multi-hop Mesh propagation	✓	✓	✓	✓	✓		
SQLite Data Sync	✓		✓			✓	
Store-Carry-Forward	✓	✓	✓	✓	✓	✓	
Structured SOS Payloads	✓	✓				✓	

Android Only	✓		✓			✓	✓
No Internet/Cell Dependency	✓	✓	✓	✓	✓	✓	✓
Algorithm Text Filtering(Jaccard /Levenshtein)	✓						

Past research shows that while phone-to-phone networks work, they lack the permanent storage and setup needed for high-stakes rescues. This study is important because it bridges those gaps by combining SQLite local phone storage with store-and-forward mesh routing. This setup ensures that SOS signals from Tagum's isolated riverside villages stay alive until they reach responders, giving them a reliable lifeline that temporary systems cannot provide.

Definition of Terms

- Android-based Communication System —The phone software used in this study. It connects devices directly to each other using Android's built-in phone-to-phone systems, allowing people to talk without any cellular service or Wi-Fi.

- Bluetooth Low Energy (BLE) —A short-range wireless system used in this study to find nearby devices. It lets phones discover each other automatically while saving as much battery power as possible.

- Data Volatility — The danger of losing emergency messages if a phone restarts, crashes, or runs out of battery before the message can safely reach rescuers.

- Decentralized Framework — A network structure where there is no central “hub” or cell tower; instead, every user’s smartphone acts as both a sender and a relay for information.

- Digital Darkness — A state during a disaster where all traditional communication infrastructures (internet, cellular signal, and electricity) are paralyzed, leaving a community technologically isolated.

- Duty-Cycling — A power-management technique where the device’s wireless radios (Wi-Fi/Bluetooth) are periodically toggled on and off to conserve battery life while maintaining the ability to discover nearby nodes.

- Golden Period – The critical window of time immediately following a disaster when medical intervention and rescue are most likely to prevent loss of life.

- Last-Mile Gap — The final, most difficult leg of communication between emergency responders and victims in physically isolated or “river-locked” areas that traditional networks cannot reach.

- Mobile Ad-hoc Network (MANET) — A spontaneous, self-configuring network of mobile devices connected wirelessly without the use of any established infrastructure or centralized administration.

- Multi-hop Propagation — The process where a message “hops” from one smartphone to another across a chain of users until it eventually reaches a responder or a functional gateway.

- No-Root Communication – A software approach that allows the application to access advanced networking features (like Wi-Fi Direct) without requiring the user to bypass the Android operating system’s security restrictions.

- Opportunistic Networking — A communication paradigm where data is transmitted over a dynamic and intermittent network; It relies on the

“opportunity” of mobile nodes coming into contact with one another to exchange data without requiring a continuous end-to-end path.

- Peer-to-Peer (P2P) – A direct communication model between two or more mobile devices that does not require an intermediary access point, server, or telecommunications provider.

- Shadow Network — The name of the proposed system; it refers to an “underground” or alternative network that operates in the shadow of the primary, failed telecommunications grid.

- Signal Loss —The weakening of wireless signals when they try to pass through blockages like heavy rain, floodwaters, or thick riverside bushes in Tagum City.

- Single Point of Failure — A critical part of a network that breaks the whole system if it fails. In this study, it refers to central cell towers that leave an entire city completely silent if they collapse during a disaster.

- SQLite Local Persistence — A built-in phone storage system that safely saves emergency SOS records directly onto the phone’s memory. This keeps the messages safe even if the phone restarts or dies.

- State of the Art — The best technology available right now. The study critiques these modern tools because they do not have the strong data structure and local storage needed during major disasters.

- Store-Carry-Forward — A message-sharing method where a phone saves an emergency text, holds onto it while the user walks or drives around, and automatically sends it out as soon as it gets close to another phone or sensor.

- Structured Payload — A standard form for emergency texts that fills in exact boxes like GPS location, signal accuracy, and what the victim needs like food or medicine. This organized format makes it much faster for rescue teams to read and process than a messy, typed-out chat message.

- Wi-Fi Direct — A wireless system that lets Android phones link directly to each other over long distances and at fast speeds. It works completely on its own without needing an internet router or cellular towers.

CHAPTER II

METHODOLOGY

The methodology employed in this research is described in this section. The Modified Waterfall Model is selected as a framework for the development of the system in this study. The Modified Waterfall Model is very systematic and organized which makes it a good fit for this study. It has a clear sequence of phases therefore it allows going back to the previous phase to resolve certain technical problems. This means that the team may verify and improve only decentralized aspects without affecting the whole life cycle and converting it into the iterative or cyclic process.

Figure 1 illustrates the use of the Modified Waterfall Model by the developers in the execution of this project, as shown below.

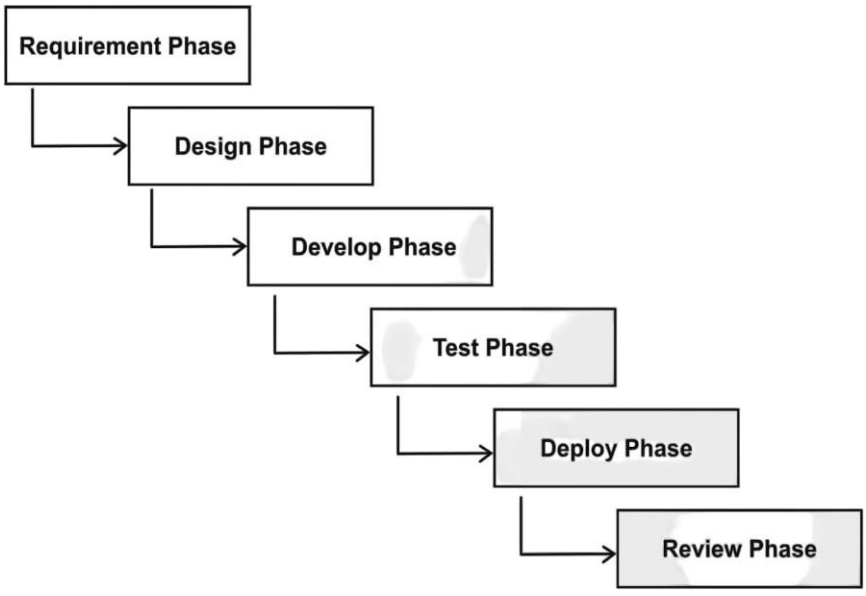


Figure 1. Modified Waterfall Model

The team concentrated on gathering and also analyzing all relevant information and technical parameters during the Requirements Phase for this decentralized communication system, creating a clear foundation before proceeding. In these part, we needed to obtain data about the last-mile communication problems in barangays along the riverside of the Tagum City, Additionally, topics of hardware issues and the technical specifications needed for a no-root Android system were also discussed during this phase in order to verify structural practicality of the system before continuing onto the next step.

During the Design Phase, specific system architectures and designs were developed according to the project's resilience needs. The proponents of this phase worked on developing the system architecture for SQLite Local Persistence database as well as Store-Carry-Forward (SCF) routing logic ensuring full definition of data structures. Additionally this phase included User Interface (UI) design for SOS sender and Offline Triage Map, therefore creating visual and logical specifications necessary to start coding.

The Development Phase involved the actual coding and building of the system in line with the approved design specifications. The front-end developers worked on the mobile interface, while the back-end developers incorporated the SQLite database and the multi-hop routing protocols in fulfilling Objective 2 and 3 of this research based on the architectural designs.

The Test Phase is where a thorough assessment of this system is performed so as to check for its performance, functionality, and dependability against all specifications. The team used this system to simulate a situation where no light is present so as to check that all components of the system work as they should, namely, the SOS sending and saving of information on the local devices. The changes to the sequential paths in the model tell us that if there are any technical glitches or problems regarding the design or the protocol, we can go back to the Development and Design Phases in order to rectify these problems through troubleshooting and fixing the code.

During the Deployment Phase, the system enters into the stage of expert reviews and initial implementation planning. Instead of a large-scale

implementation in the field, the implementation is evaluated in the technical review performed by one expert from the field of telecommunications or network engineering who observes the stability of Android-to-Android links and the quality of decentralized data synchronization and ensures proper work of the system.

The Review Phase is when the system conducts a concluding evaluation and its operation checks against the known technical standards set down. Here, an in-depth examination of all system logs takes place to check for performance issues such as trends in signal failure or losses and problems with traffic redirection. Where minor non-compliance with the standards is detected, the previously planned controlled tracing back mechanism could be applied so that the authors would be able to revert only to the Development Phase and apply the necessary software patch or upgrade the routing module. This way, the system's performance would have been assured to be thoroughly tested and stabilized before the launch.

System Planning

At the early stage of planning, the project proponents have formed the basic aims of the project and devised an in-depth scheme. This stage was a very important organizational element that included Project Organization Team, the Work Breakdown Structure (WBS), and constructing a Gantt chart to monitor the technical milestones. The abovementioned steps made it possible to ensure that all the project phases met the overall timeline for creating a resilient communication system.

The system's proponents detailed its potential technical elements, data requirements, and approaches for construction and implementation. Discussions were held to find efficient strategies to deal with eventual challenges. The developers defined the objectives of the project and set out the important conditions for the development of the system. In addition, the creators of this phase documented a large number of materials for the design stage by collecting all the important data and materials when working with similar models.

Project Team Organization

The project team's name for the system is Shadow Network: Peer-to-Peer Disaster Messaging System. The system in this section explains the project framework, where various roles and responsibilities of the project partners are specified. The structure of the project partners is presented in Figure 2 in the form of a hierarchy chart, so that readers can understand what everyone is doing.

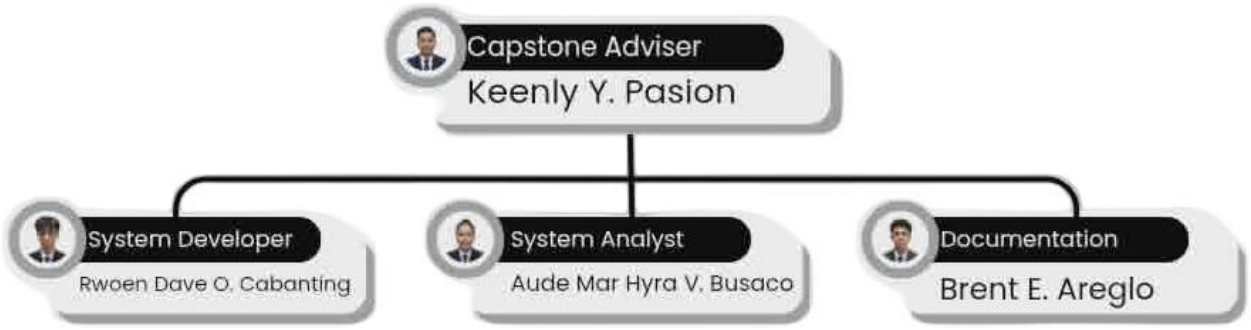


Figure 2. Project Team Organizational Chart

System Analyst supervises the development procedure by collecting technical specifications to confirm that the network satisfies disaster response standards. They assign duties for composing the system logic and ensuring that the peer-to-peer architecture conforms to stakeholder requirements. Furthermore, they ensure that the project's scope and objectives are met technically throughout its lifecycle.

Documentation is responsible for having all documents, technical details, and research materials up to date and accurate. A significant part of their work includes writing the manuscript that explains complicated networking ideas and research outcomes. It implies that he/she should evaluate the project's design and scope.

System Developer is primarily responsible for the creation and implementation of software, including user interface and main communication protocols. His/her job covers the whole technical cycle from creating the Store-Carry-Forward logic up to database integration. The Developer also conducts

thorough tests and maintenance in order to keep application functioning throughout the full collapse of infrastructure.

Work Breakdown Structure (WBS)

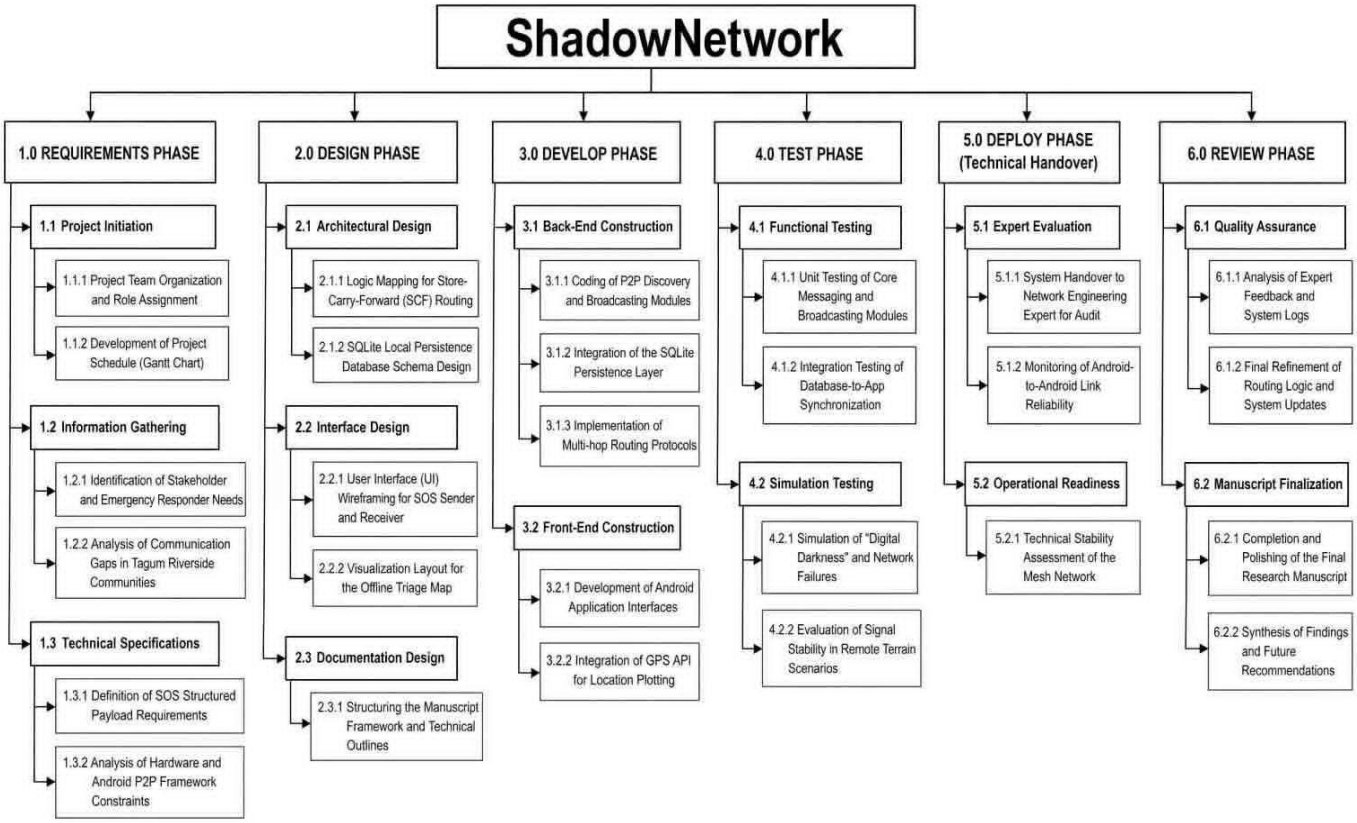


Figure 3. Work Breakdown Structure

Work Breakdown Structure (WBS), portrayed in Figure 3, is the prime foundation for the management of the project since it organizes the total scope of work into manageable functional units. The use of the WBS aligns with the Modified Waterfall Model since proponents can group every technical requirement in order from peer-to-peer (P2P) discovery to the write-up of the manuscript. This manner of design also makes it easier to assign responsibilities to the System Analyst, System Developer, and Documentation Specialist.

Gantt Chart

ShadowNetwork

GANTT CHART

2026

WBS	TASK / ACTIVITY	2026 Q1	Q2	Q3	Q4	Q5	Q6
1.0	**REQUIREMENTS PHASE**						
1.1	**Project Initiation**						
1	1.1.1 Project Team Organization and Role Assignment						
	1.1.2 Development of Project Schedule (Gantt Chart)						
1.2	**Information Gathering**						
	1.2.1 Identification of Stakeholder and Emergency Responder Needs						
	1.2.2 Analysis of Communication Gaps in Tagum Riverside Communities						
1.3	**Technical Specifications**						
	1.3.1 Definition of SOS Structured Payload Requirements						
	1.3.2 Analysis of Hardware and Android P2P Framework Constraints						
2.0	**DESIGN PHASE**						
2.1	**Architectural Design**						
	2.1.1 Logic Mapping for Store-Carry-Forward (SCF) Routing						
	2.1.2 SQLite Local Persistence Database Schema Design						
2.2	**Interface Design**						
	2.2.1 User Interface (UI) Wireframing for SOS Sender and Receiver						
	2.2.2 Visualization Layout for the Offline Triage Map						
2.3	**Documentation Design**						
	1.3.1 Structuring the Manuscript Framework and Technical Outlines						
3.0	**DEVELOP PHASE**						
2.1	**Back-End Construction**						
	3.1.1 Coding of P2P Discovery and Broadcasting Modules						
	3.1.2 Integration of the SQLite Persistence Layer						
	3.1.3 Implementation of Multi-hop Routing Protocols						
3.2	**Front-End Construction**						
	2.2.1 Development of Android Application Interfaces						
	3.2.2 Integration of GPS API for Location Plotting						
4.0	**TEST PHASE**						
4.1	**Functional Testing**						
	4.1.1 Unit Testing of Core Messaging and Broadcasting Modules						
	4.1.2 Integration Testing of Database-to-App Synchronization						
4.2	**Simulation Testing**						
	4.2.1 Simulation of "Digital Darkness" and Network Failures						
	4.2.2 Evaluation of Signal Stability in Remote Terrain Scenarios						
5.0	**DEPLOY PHASE (Technical Handover)**						
5.1	**Expert Evaluation**						
	5.1.1 System Handover to Network Engineering Expert for Audit						
	5.1.2 Monitoring of Android-to-Android Link Reliability						
5.2	**Operational Readiness**						
	5.2.1 Technical Stability Assessment of the Mesh Network						
6.0	**REVIEW PHASE**						
6.1	**Quality Assurance**						
	6.1.1 Analysis of Expert Feedback and System Logs						
	6.1.2 Final Refinement of Routing Logic and System Updates						
6.2	**Manuscript Finalization**						
	6.2.1 Completion and Polishing of the Final Research Manuscript						
	6.2.2 Synthesis of Findings and Future Recommendations						

= Planned Duration

Figure 4. Gantt Chart

Latest completed milestone: November 11, 2026

Figure 4 shows a Gantt chart that was used for the Shadow Network creation process. It is an illustrated schedule that shows a timeline for the project including duration estimates and resource assignments for each stage of the Modified Waterfall Model. Additionally, the Gantt Chart serves as an important tool used to identify critical milestones such as the minimum time needed for the integration of the protocols and the dependencies among the tasks of P2P module creation and the system-wide testing stages.

System Analysis

Based on the initial study of disaster response in Tagum City, the developers discovered that both riverside residents and rescue teams face difficult situation due to breakdown of cellular communication networks. The goal of the Shadow Network is connectivity for the last mile. This is accomplished through the use of SQLite Local Persistence for dependable storage and Store-Carry-Forward technology to permit multiple routing. Another aspect of this system is keeping data readily accessible for rescuers through it’s online mapping feature.

System Architecture

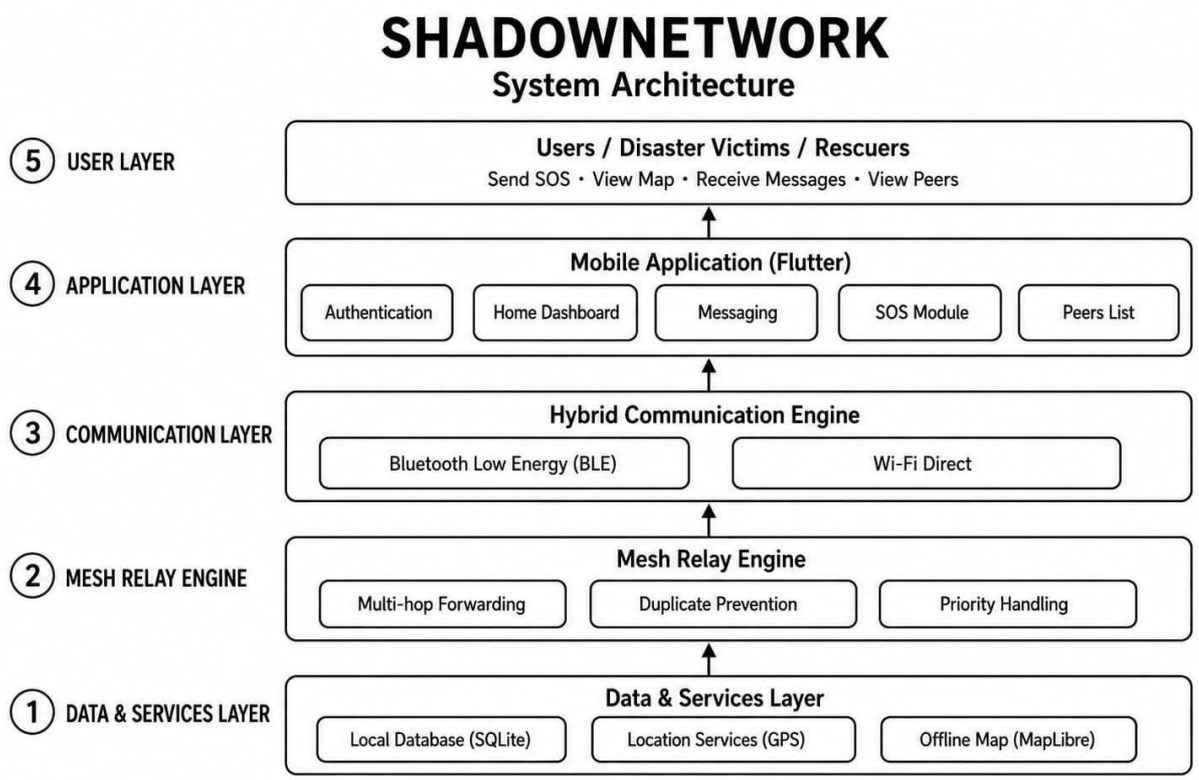


Figure 5. System Architecture

The Shadow Network system architecture exhibits a five-layer modular framework depicted in Figure 5, aiming at achieving uninterrupted communication irrespective of the collapse of the infrastructure. In this layer, the Data & Services Layer employs SQLite for local data storage and GPS technology for creating SOS messages that are location specific. The core process of Mesh Relay Engine and Hybrid Communication Engine handle decentralized routing and peer discovery by utilizing Bluetooth Low Energy (BLE) and Wi-Fi Direct. The Flutter-based Application layer abstracts the underlying core various processes by providing users with necessary user-friendly interface for performing SOS messaging and offline map visualization. The uniqueness of the architecture lies in the fact that all these processes combined make it impossible to disrupt communication in spite of the fact that there is no mobile network available.

Conceptual Framework

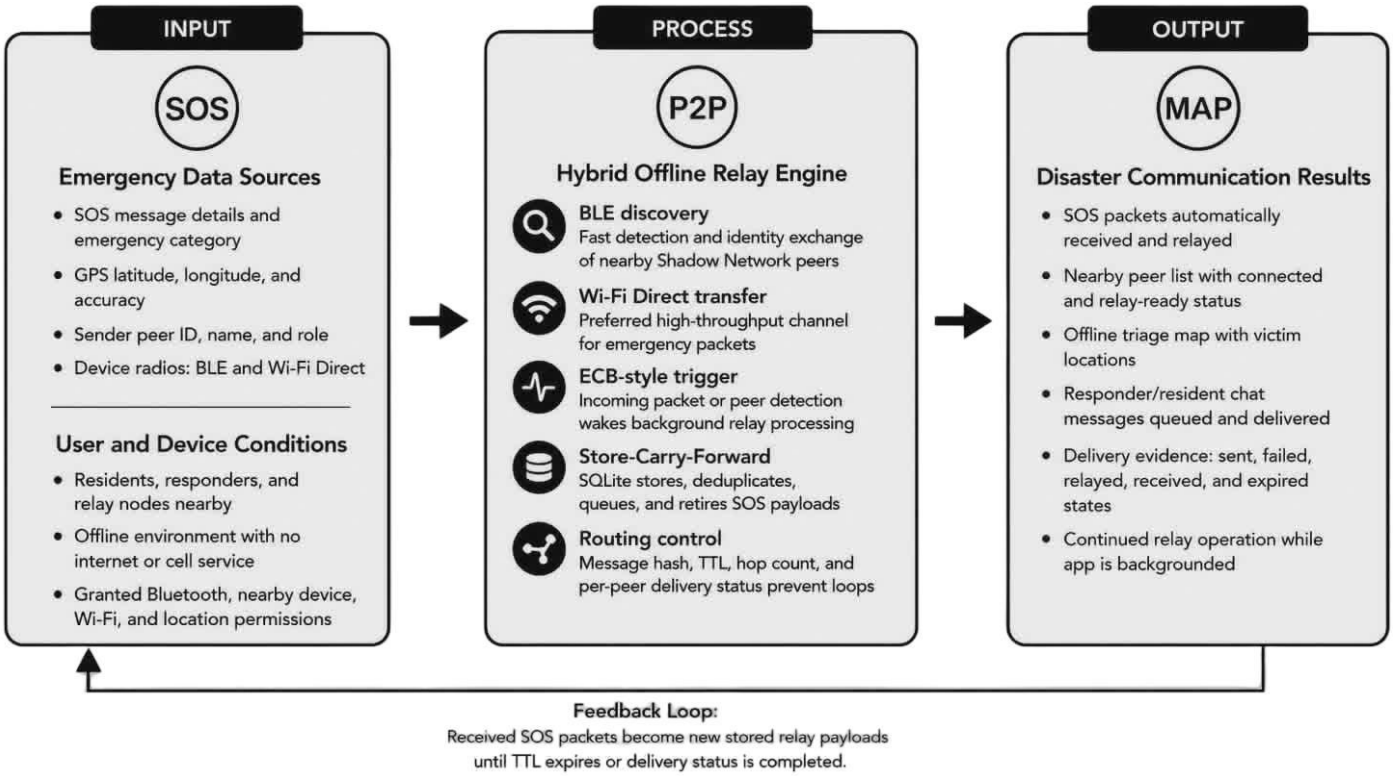


Figure 6. Conceptual Framework

The model displayed in Figure 6 adopts an Input-Process-Output approach in addition to the incorporation of a feedback loop so that communication will go on uninterrupted through instances of total network blackout. The input stage encompasses various kinds of significant information like GPS data, triage categories, and peer profiles in a totally offline mode. Process stage employs Hybrid Offline Relay Engine in executing BLE discovery and Wi-Fi Direct transmitting data and using automated ECB background trigger through SCF SQLite logic and routing control. As a result, Output entails working disaster response tools in the form of active peer visibility, background synchronization, and online coordinate plotting of the offline triage map. Ultimately, Feedback Loop transforms successfully received outputs into localized inputs for the next hops of the device, ensuring that multi-hop routing continues without interruption.

Functional and Non-Functional Requirements

Presented below are the anticipated functionality requirements and performance characteristics pertaining to the Shadow Network: Mobile App-Based Peer-to-Peer Disaster Messaging System:

Functional Requirements

1. To implement an Emergency Cell Broadcast (ECB) mechanism where incoming SOS packets automatically transfer to the system background for processing the moment the device scans or detects another peer's message, bypassing manual connection approvals.

2. To implement a One-Touch SOS broadcast feature that automatically packages the user’s GPS coordinates and timestamps into a structured data packet for immediate transmission.

3. To enable peer-to-peer (P2P) discovery and handshaking using Bluetooth Low Energy (BLE) and Wi-Fi Direct, allowing devices to link without cellular towers.

4. To utilize Store-Carry-Forward (SCF) logic, allowing the system to save incoming messages from other peers and relay them once a new device comes within range.

5. To integrate a local persistence layer using SQLite to store all incoming and outgoing messages, ensuring data is not lost if the application is closed or the device loses power.

6. To display a real-time Triage Map using offline map tiles, plotting the locations of reported victims and distress signals for responders to visualize.

7. To facilitate private text messaging between discovered peers within the mesh network, allowing for coordination between residents and rescue teams.

8. To automate the background scanning process, ensuring the device constantly looks for new nodes to expand the mesh coverage without manual user intervention.

9. To implement a decentralized dashboard l that will display connections to peers, the quality of the signal, and logs of any emergency data received or transmitted.

Non-Functional Requirements

- **Availability:** The system must operate in Digital Darkness scenarios where all facilities of external telecommunications (LTE, 5G, Fiber) are denied.

- **Usability:** The interface should be designed in a user-friendly manner and be aligned with the high-stress character of catastrophic events.

- **Reliability:** The message sending protocol should ensure that the message is unique and cannot be duplicated in the message flow.

- **Scalability:** The system should be able to transfer a message using increasing hops.

- **Efficiency:** The solution must minimize energy consumption through the use of BLE to perform background scanning of the environment and activate Wi-Fi Direct only for data transfers

- **Performance:** Single-hop payload delivery and peer discovery must take no longer than 3 to 10 seconds at a distance of 5 to 15 meters in the same room, while in buildings, the duration must be in the range of 10 to 30 seconds at 10 to 25 meters distance and 30 to 70 meters distance in the open line-of-sight. Multi-hop communication using 2 to 3 relay phones must send messages in 1 to 5 minutes over a distance of 30 to 100 meters, while the Store-Carry-Forward relaying method must deliver the payload in 5 minutes to several hours depending on the movement of the nodes.

In this section, the particular functional capacities and performance requirements that are needed for the smooth functioning of the Shadow Network in places without communications are defined. The major technical specifications are presented here.

Table 2. Software Requirements

Software	Use
VS Code	The primary Integrated Development Environments (IDEs) used for coding and compiling the Flutter application.
Flutter SDK	The cross-platform framework used to develop the application's user interface and core logic.
SQLite	The serverless, local database engine used for storing persistent SOS messages and peer data on the device.

Android SDK (API 21+)	Required to access system-level hardware features like Wi-Fi Direct (P2P) and Bluetooth Low Energy (BLE).
MabLibre	The open-source mapping provider used to generate offline map tiles for the triage map.

Table 3. Hardware Requirements

Hardware	Recommended Specifications
Development Laptop/PC	Windows 10/11, 8GB RAM (minimum), Intel Core i5 or equivalent for running Android emulators.
Primary Android Device	Android 5.0 or higher with built-in Wi-Fi Direct and BLE support for P2P networking.
Secondary Test Nodes	At least 2-3 additional Android smartphones to simulate and test the multi-hop Store-Carry-Forward logic.
Internal Storage	At least 100MB of free space on mobile devices for local database storage and offline map caching.
GPS Module	Integrated mobile GPS hardware for real-time coordinate fetching and SOS payload tagging.

Use Case Diagram

This section of the paper describes system user-to-system interaction indicating how people and various parts of the system operate through the application. It provides a detailed idea about the features and processes involved in operation of the system from the user's perspective and the underlying technologies of decentralized communication protocols.

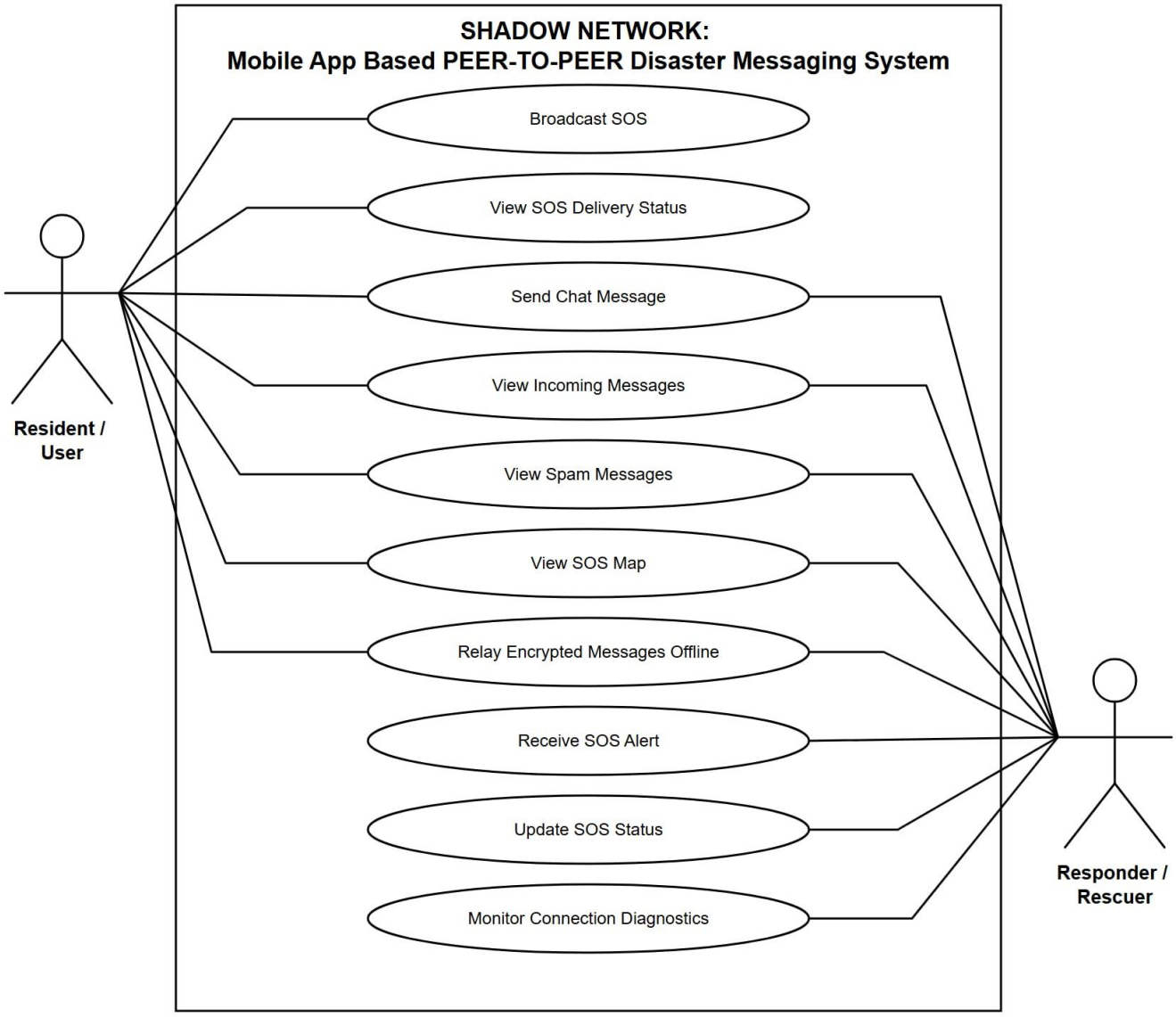


Figure 6. Shadow Network Use Case Diagram

The diagram depicted in figure 6 illustrates the Use Case Diagram associated with the Shadow Network in which all the interactions between two actors, namely the Resident/User and Responder/Rescuer are shown. The diagram is concentrated on the goals oriented towards the users, while the solving and technical operations in the form of Bluetooth Low Energy discovery, Wi-Fi Direct transmission, SQLite persistence, message moderation, and Store-Carry-Forward processing are presented as internal mechanisms of the system rather than as separate actors or use cases.

Broadcast SOS - As the name suggests, allows the Resident/User to create his/her emergency message containing the details regarding category, description, GPS coordinates, location accuracy, recipient, time stamp, and routing details.

View SOS Delivery Status - This use case makes it possible for the Resident/User to see the status of his/her SOS message such as pending, queued, relayed, delivered, etc. Here the displayed information can be updated according to the information stored in the system.

Send Chat Message - When it comes to this use case, the Resident/User and Responder/Rescuer are enabled to send and receive messages from each other in order to coordinate their actions.

View Incoming Messages - Finally, this use case allows both actors to receive messages, and any SOS alert and chat messages intended for their device. The received message being sent through intermediary devices will not be visible unless the intended recipient receives them.

Viewing spam messages - This function allows both users different information forms of spam messages to be accessed by the users remotely. The system works on eliminating the ambiguous messages from the system feed, leaving them in the local storage, where they will be checked and verified.

Viewing SOS map - This function permits both users to view the past points of sending real SOS messages on the map. The spam messages will not be displayed on the map, i.e. the system will not show spam messages on the map points.

Relay Encrypted Messages Offline - This function allows both users to relay messages offline through users/residents' devices as the SOS messages sent during encryption may be kept and relayed through all the devices available in order to deliver them to the respective recipients. Transporting devices have no access to the message content itself.

Receiving SOS alert - This function allows responders to receive SOS alerts sent to the responder's registered device. SOS alerts contain all needed data about the emergency event, its type, description, sender, location, and delivery methods.

Updating SOS status - This function allows responders to update the status of the SOS request. The status is sent through the existing connection to the user/resident by means of the peer-to-peer connection.

Diagnostics - This function allows the responder to gain information about the operation of the systems: discovered devices, strength of signals, connections, relay attempts, and other parameters needed for diagnostics.

Context Flow Diagram

This section describes the system and its interaction with other external parties, as well as how the data travels between Shadow Network and users in the disaster area.

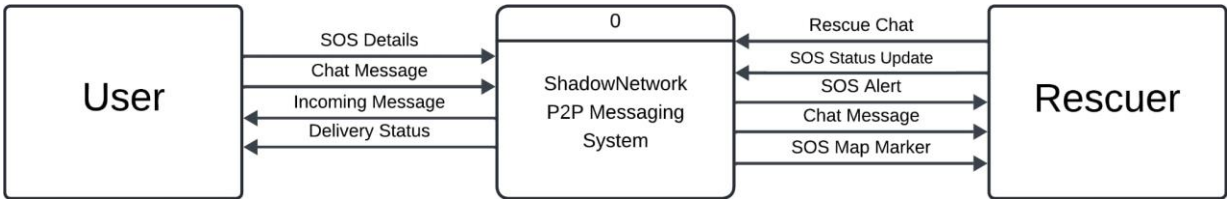


Figure 7. Context Flow Diagram

The Context Data Flow Diagram for the Shadow Network P2P Messaging Platform is shown in Figure 7 and illustrates how the platform interacts with external entities such as the User and Rescuer. The User is capable of sending SOS information and chat messages to the platform while receiving messages and information updates. The Rescuer receives SOS messages, chat messages, and SOS map indications and is capable of sending response messages. The diagram portrays user and rescue peer searching, encrypted data transmission, and store-carry-and-forward functions as internal activities of the platform instead of external entities that can be showed in the context.

Data Flow Diagram

This section elaborates the processes of the system, data acquisition, and data storage compared to the Context Flow Diagram. It breaks down the system into smaller components and explains how the data travels through the processes of the system and its local storage.

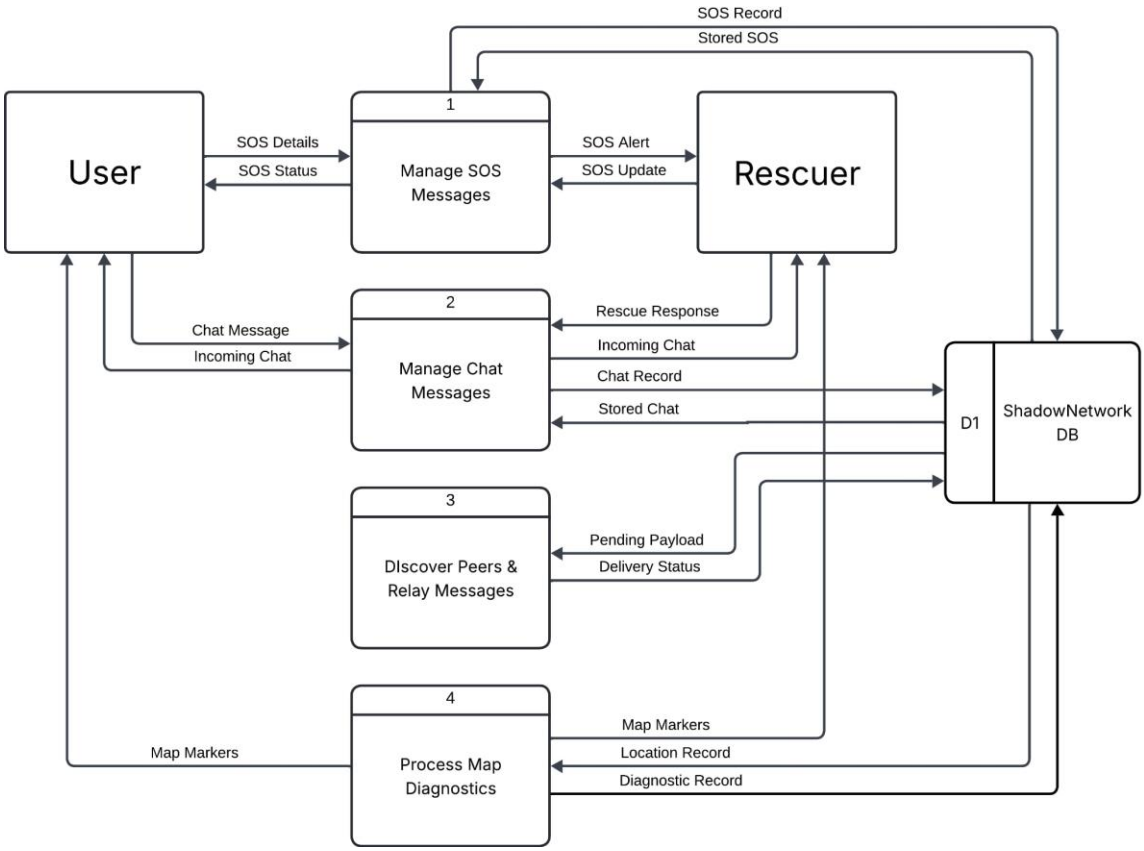


Figure 8. Data Flow Diagram

Figure 8 shows the various internal sub processes in the system, which include Manage Chat and SOS Messaging, Store-Carry-Forward (SCF) Relay Service, and Process Map and Connection Diagnostics. The first sub process, Manage Chat and SOS Messaging gives a standardized bidirectional communication interface through which it logs the user and responder messaging events both as relational data in SQLite Local Storage and gives back the Incoming Chats and alerts and Map Markers to the client. The Store-Carry-Forward Relay Service on the other hand involves the managing of the ad-hoc queuing and the

transfer of the Outgoing and Incoming payloads with some neighboring Peer Nodes, whereby carrying out dynamic updates concerning the tracking of hop made and the acknowledgement status for delivery in the database. Lastly, Process Map and connection diagnostics gets the recorded GPS Coordinates and Connection Stats from the SQLite database to give accurate SOS Map Markers and connection diagnostic measurements to the rescuer.

System Design

The design section of the Shadow Network establishes the core architecture, data structures, and operational interfaces. During this stage, the requirements of the analysis phase are transformed into feasible technical or design specifications which will serve as the road map that lays down the groundwork for the operation of the system during emergency situations. The ERD in figure 9 illustrates the structure of the data base schema indicating the linkage of the various data elements within it. The establishment of all these logical links is crucial to ensure that the offline database is well-structured so that it consistently receives and stores SOS payloads at the nodes of the network.

Entity Relationship Diagram (ERD)

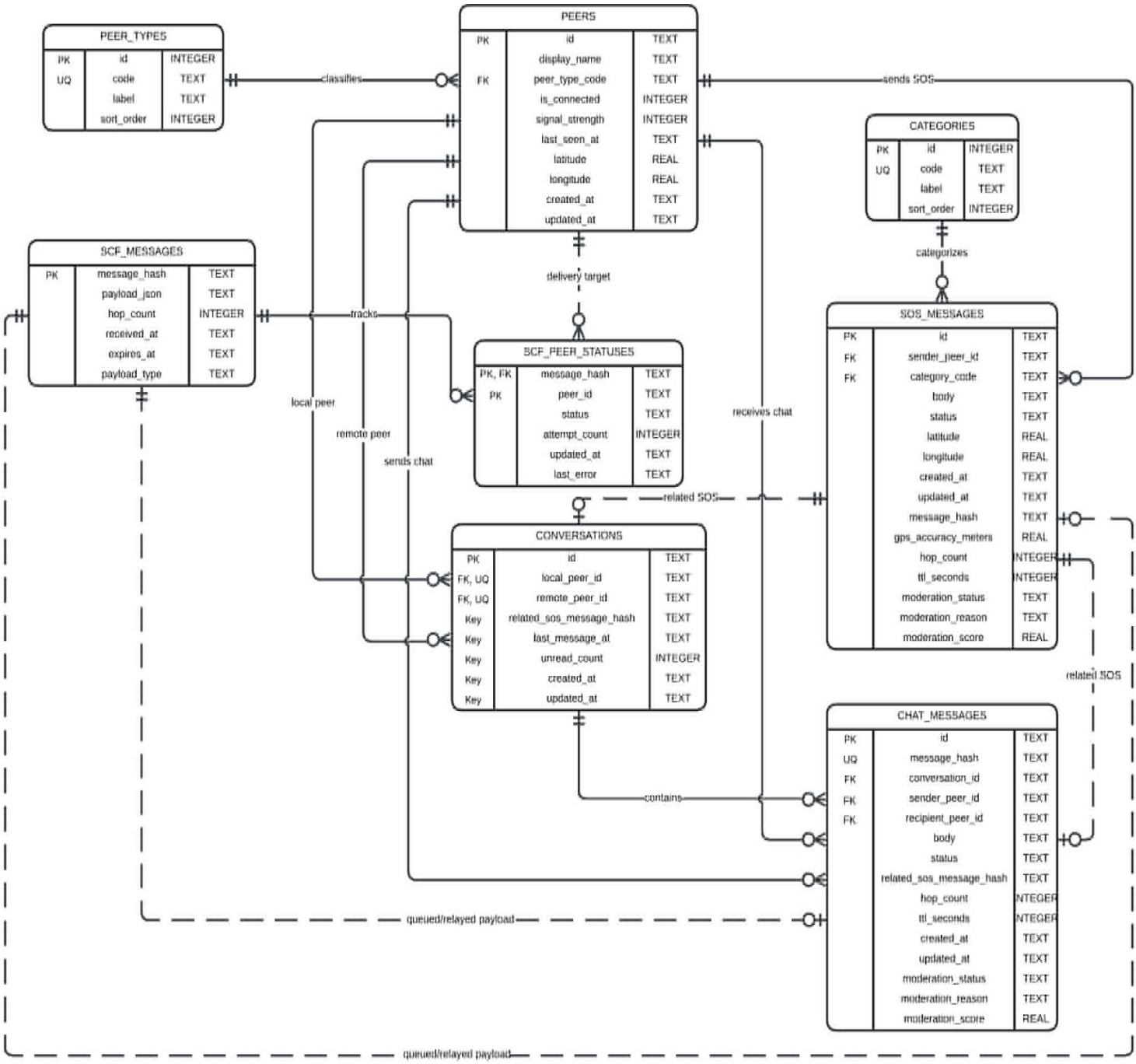


Figure 9. Entity Relationship Diagram (ERD)

The ERD provides the abstraction used to visualize how the various main tables within the database relate to each other. As demonstrated by the schema, all entities depicted indicate how the various information such as distress messages from users, peer informatics, and state of message transmission is stored locally. The emphasis of this schema is on the messages table since it generates its identification for messages using a message_hash and it also contains other fields such as hop count and time-to-live. All these relationships are established in order to create a reliable database structure that assures the consistency of data, and thus enables the system to conduct fast queries during emergencies.

JSON Schema Diagram

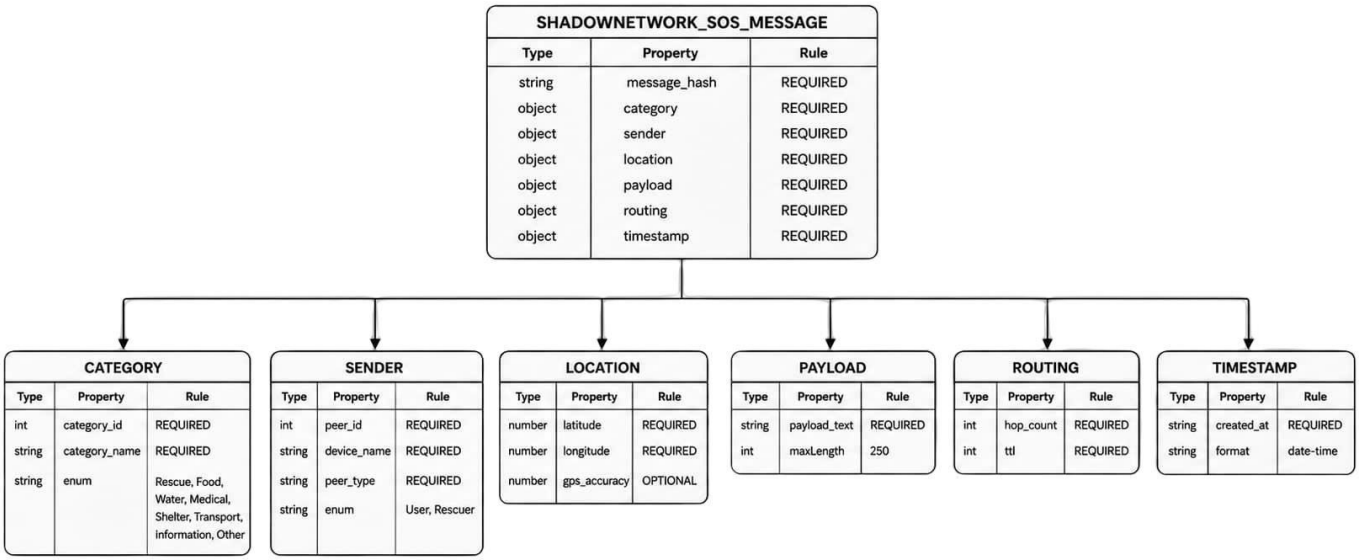


Figure 10. JSON Schema Diagram

The JSON schema shown in Figure 10 applications the structure, which was designed to establish a common framework for the information shared in the decentralized network. The overall structure followed the hierarchy, making Location, Routing, and Timestamp information become separate data objects,

facilitating the light-weight messaging content of the distress signals to allow low-bandwidth transfer, while being packed with metadata. By means of message_hash, data can be eliminated as duplicate entries in the local database, maintaining the high efficiency of the local storage, despite multiple signals being sent by several peers. The attributes relating to routing, including hop_count and ttl, contain the rules ensuring the normal operation of the Store-Carry-Forward scheme for the management of information distribution and providing network congestion.

Data Dictionary

In this section, the metadata in the database is shown, including the naming of fields, types of data, and description of the fields which are applied in the database itself. It is a good guidance for understanding the internal format of the structure, ensuring the standard way of storing the information related to emergency cases.

Table 4. Data Dictionary

Table Name	**Description**
SOS_MESSAGES	Stores the core content, location data, and metadata.
PEERS	Tracks unique devices discovered within the mesh network.
CHAT_MESSAGES	Archives peer-to-peer text conversation feeds passing through active mesh lanes.
CONVERSATIONS	Manages unique session threads created between local and discovered remote peers.

CATEGORIES	Defines the types of emergencies.
PEER_TYPES	Categorizes nodes.
SCF_MESSAGES	Stores raw JSON network packet structures for intermediate Store-Carry-Forward handling.
SCF_PEER_STATUS	Tracks packet relay synchronization histories across physical peer endpoints.

Table 5. SOS_MESSAGES

Table Name	Data Type	Constraints	Description
id	TEXT	Primary Key	Unique identifier for the SOS message.
sender_peer_id	TEXT	Foreign Key	ID of the peer who sent the message.
body	TEXT	Nullable	Text details of the emergency.
category_code	TEXT	Foreign Key	Code linking to the emergency category.
status	TEXT	Not Null	Lifecycle status of the alert.
latitude	REAL	Not Null	GPS latitude coordinate of the sender.
longitude	REAL	Not Null	GPS longitude coordinate of the sender.

created_at	TEXT	Not Null	Timestamp when the message was created.
updated_at	TEXT	Not Null	Timestamp of the last record update.
message_hash	TEXT	Not Null	Unique hash used to drop duplicate packets.
gps_accuracy_meters	REAL	Nullable	Accuracy margin of the GPS coordinates.
hop_count	INTEGER	Not Null	Number of times the message was relayed.
ttl_seconds	INTEGER	Not Null	Time-to-Live expiration limit in seconds.
moderation_status	TEXT	Not Null	Reclassified to 'SPAM' if text similarity rules were violated.
moderation_reason	TEXT	Not Null	Records if Jaccard Similarity or Levenshtein Distance triggered the spam flag.
moderation_score	REAL	Not Null	Calculated textual redundancy coefficient

Table 6. PEERS

Table Name	Data Type	Constraints	Description
id	TEXT	Primary Key	Unique hardware or network ID of the device.
display_name	TEXT	Nullable	Optional user nickname for the node.
peer_type_code	TEXT	Foreign Key	Code linking to the peer classification role.
is_connected	INTEGER	Not Null	Connection state indicator (1 = True, 0 = False).
is_connected	INTEGER	Nullable	RSSI connection signal strength in dBm.
last_seen_at	TEXT	Not Null	Timestamp of the last verified contact.
latitude	REAL	Nullable	Last known latitude of the peer hardware.
longitude	REAL	Nullable	Last known longitude of the peer hardware.
created_at	TEXT	Not Null	Timestamp when the peer was first discovered.
updated_at	TEXT	Not Null	Timestamp of the last metadata modification.

Table 7. CHAT_MESSAGES

Table Name	Data Type	Constraints	Description
id	TEXT	Primary Key	Unique identifier for the chat entry.
message_hash	TEXT	Unique	Unique hash used to eliminate duplicate chats.
conversation_id	TEXT	Foreign Key	ID linking the message to a specific thread.
sender_peer_id	TEXT	Foreign Key	Unique ID of the transmitting node.
recipient_peer_id	TEXT	Foreign Key	Unique ID of the receiving target node.
body	TEXT	Not Null	Plaintext content of the chat message.
status	TEXT	Not Null	Transmission state (e.g., Sent, Delivered).
related_sos_message_hash	TEXT	Nullable	Optional link connecting chat to an active SOS.
hop_count	INTEGER	Not Null	Total mesh relay steps traveled by the packet.

ttl_seconds	INTEGER	Not Null	Storage expiration timeout threshold in seconds.
created_at	TEXT	Not Null	Timestamp when the text message was created.
updated_at	TEXT	Not Null	Timestamp of the last status change.
moderation_status	TEXT	Not Null	Automatically set to “SPAM” if text similarity matched existing entries.
moderation_reason	TEXT	Not Null	Logs if Jaccard or Levenshtein background algorithms flagged text repetition.
moderation_score	REAL	Not Null	Metric tracking text duplication values from 0.0 to 0.1

Table 8. CONVERSATIONS

Table Name	Data Type	Constraints	Description
id	TEXT	Primary Key	Unique identifier for the chat room session.
local_peer_id	TEXT	Foreign Key, Unique	Hardware reference ID of the local host device.
remote_peer_id	TEXT	Foreign Key, Unique	Hardware reference ID of the remote chat target.
related_sos_message_hash	TEXT	Nullable	Links the conversation to a specific emergency.
last_message_at	TEXT	Nullable	Timestamp of the most recent chat activity.
unread_count	INTEGER	Not Null	Quantitative count of unread incoming messages.
created_at	TEXT	Not Null	Timestamp when the thread was opened.
updated_at	TEXT	Not Null	Timestamp of the last table update.

Table 9. CATEGORIES

Table Name	Data Type	Constraints	Description
id	INTEGER	Primary Key, Auto Increment	Unique internal ID for database optimization.
code	TEXT	Unique, Not Null	System identifier code (e.g., 'MED', 'FIRE').
label	TEXT	Not Null	Plaintext name displayed on user screens.
sort_order	INTEGER	Nullable	Numeric value used for ordering lists on UI.

Table 10. PEER_TYPES

Table Name	Data Type	Constraints	Description
id	INTEGER	Primary Key, Auto Increment	Unique internal identifier for the node role.
code	TEXT	Unique, Not Null	Role configuration string (e.g., 'VCTR', 'RSCR').
label	TEXT	Not Null	Display description of the role (e.g., Victim).
sort_order	INTEGER	Nullable	Numeric value used for sorting user lists.

Table 11. SCF_MESSAGES

Table Name	Data Type	Constraints	Description
message_hash	TEXT	Primary Key	Target reference hash identifying the cached item.

payload_json	TEXT	Not Null	Raw serialized JSON string of the mesh payload.
hop_count	INTEGER	Not Null	Total distribution steps across active data links.
received_at	TEXT	Not Null	Timestamp when the device ingested the packet.
expires_at	TEXT	Not Null	Threshold timestamp for automated sweep removal.
payload_type	TEXT	Not Null	Categorizer type identifier (e.g., 'SOS', 'CHAT').

Table 12. SCF_PEER_STATUS

Table Name	**Data Type**	**Constraints**	**Description**
message_hash	TEXT	Primary Key, Foreign Key	Identifies the managed packet object hash.
peer_id	TEXT	Primary Key, Foreign Key	Identifies the target neighbor node for matching sync passes.
status	TEXT	Not Null	Synchronize outcome status tracker (e.g., Synced).
attempt_count	INTEGER	Not Null	Count of distribution delivery attempts made.

updated_at	TEXT	Not Null	Timestamp logging network connection events.
last_error	TEXT	Nullable	Description or error log caught during transfer failure.

Technologies, Concepts, and Theories

This part gives details on how the Shadow Network functions. The first phase involves gaining user information by storing SOS alerts and location data using GPS technology. The data collected is then organized in standard format and associated with a unique ID, which prevents duplication. The next phase of the operations is sharing such information using Send-Carry-Forward technique whereby phones store alerts and pass on messages when they detect other devices within proximity. Then, the Shadow Network sorts and categorizes alerts by type and presents their respective locations using offline maps.

To build this system, we utilized the available technologies in the market. For example, we used Flutter to build user-friendly applications for iPhones and Android devices. To provide offline storage capabilities, we used SQLite for offline storage. The devices in the Shadow Network communicate using Wi-Fi Direct and Bluetooth Low Energy (BLE) without needing mobile cellular connections. With respect to mapping, we deployed OpenStreetMap for the rescuers to have offline maps for better operation in disaster areas. Furthermore, the coding process was done inside Visual Studio Code to ensure perfect operation and service delivery.

System Testing and Implementations

Testing of the system will be done in a thorough manner so that we can be sure that by the time someone uses it for real, it’s working just as well as expected. The tests will check three key aspects of the system: the number of messages that

complete, speed of phone data sharing, and amount of battery energy used up while using the app. The testing will cover all minor components of the system, making sure that app records GPS data properly, stores the data in a safe way and allows mobile devices to communicate properly. All testing results will be presented in an easy-to-read table showing how ready the system is.

We will conduct detailed testing on our system to see just how far we can push it. We will test our Store-Carey-Forward system in high-density areas where numerous phones are engaged in data transmission simultaneously. We will also simulate different kinds of signal interference to determine how far Bluetooth and Wi-Fi Direct will function in actual disaster areas. By timing the delivery of SOS messages from victim to rescuer through a series of phones, we will demonstrate that the app is reliable in real-world situations. In addition, we will conduct extensive testing to assure that our app will not lose data even when phones are experiencing weak battery or signal problems. When we make the app available to rescue workers, we will make it ready for immediate utilization. It will already contain detailed maps and critical data pre-installed so that internet access is not needed. Lastly, we will evaluate our success based on how easily the app can be used in emergencies and how quickly phones are located in the field.

System Maintenance

We are preparing a plan for keeping our system efficient and secure in the future. With this plan, we will be able to make sure how to keep text messages secure and convenient for users by defining the authenticated persons using the account. This plan will offer anonymity and protection of the connections of the people using the services from any possible attack from hackers. The well-established security system prevents criminals from sending fake messages and hacking unsecured connections by using decentralized identification of the users involved. In addition, the algorithm will enhance the process by ensuring the testing of every incoming message just before its storage in the database.

To make sure that the application works properly for many years to come, it will be necessary to devise a maintenance plan that will eliminate any issues regarding data functionality and help the application perform well on the new generations of mobile devices. The maintenance plan will include regular cleaning and structuring of the database, which will help to find victims' locations in real time, as the number of messages keeps rising. Also, regular updates of the offline maps and the software installed are required to make them compatible with the system of updates for Android devices.

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